

BCAC 0026: ANALYSIS OF ALGORITHM AND DATA STRUCTURES

Objective:

Credits: 04

L-T-P: 4-0-0

Module No.	Content	Teaching Hours
I	Introduction: Basic Design and Analysis techniques of Algorithms, time and space complexity, Correctness of Algorithm. Algorithm Design Techniques: Iterative techniques, Divide and Conquer, Dynamic Programming, Greedy Algorithms. Arrays: Single and Multi-dimensional Arrays, Sparse Matrices. Searching Techniques and Complexity Analysis: Linear and Binary search, Medians & Order Statistics. Stacks and Queues: Implementing stack using array and linked list, Prefix, Infix and Postfix expressions, Utility and conversion of these expressions from one to another; Array and Linked representation of Queue, De-queue, Priority Queues.	30
II	Sorting Techniques: Elementary sorting techniques-Bubble Sort, Insertion Sort, Merge Sort, Advanced Sorting techniques - Heap Sort, Quick Sort, Sorting in Linear Time-Bucket Sort, Radix Sort and Count Sort. Linked Lists: Singly, Doubly and Circular Lists, representation of Stack and Queue as Linked Lists. Recursion: Developing Recursive Definition of Simple Problems and their implementation; Advantages and Limitations of Recursion. Trees: Introduction to Tree as a data structure; Binary Trees, Binary Search Tree, (Creation, and Traversals of Binary Search Trees).	30

Text Book:

- Cormen T.H., Leiserson Charles E., Rivest Ronald L., Stein Clifford, Introduction to Algorithms, PHI Learning Pvt. Ltd., 2009, 3rd Edition.

Reference Books:

- Basse Sara & A.V. Gelder, Computer Algorithm: Introduction to Design and Analysis, Pearson, 2000, 3rd Edition.
- Drozdek Adam, "Data Structures and algorithm in C++", Cengage Learning, 2012, Third Edition.
- Tenenbaum Aaron M., Augenstein Moshe J., Langsam Yedidyah, "Data Structures Using C and C++", PHI, 2009, Second edition.
- Kruse Robert L., "Data Structures and Program Design in C++", Pearson.

Suggestive digital platforms web links or online course

- <https://www.oercommons.org/authoring/14873-data-structure/view>
- <https://www.oercommons.org/courses/data-structure-and-algorithms>
- https://onlinecourses.swayam2.ac.in/cec19_cs04/preview (**online course**)

Outcome: A student who successfully completes the course will have the ability to:

- CO1: Understand that various problem solving categories exist such as; iterative technique, divide and conquer, dynamic programming, greedy algorithms, and understand various searching and sorting algorithms.
- CO2: Employ a deep knowledge of various data structures when constructing a program.
- CO3: Design and construct simple object-oriented software with an appreciation for data abstraction and information hiding.

- CO4: Effectively use software development tools including libraries, compilers, editors, linkers and debuggers to write and troubleshoot programs.